Y-88 Energy Calibration, Gain Curves, and Trigger Thresholds

n_TOF EAR2 X17 — Y-88 source scan (224476–79) + long run 224503

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analysis and slides prepared with Claude (Anthropic AI assistant)

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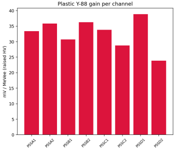
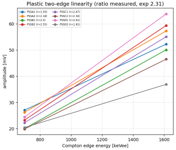
What we did

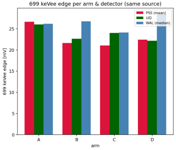
- **Y-88 source scan** (224476–79, one source per arm): an absolute energy calibration from the **699 & 1612 keVee Compton edges**, on plastics, liquids, and SiPM walls.
- Reprocessed with the LIQ PSA UserInput (the official files have no LIQ).
- Anchored the two **plastic HV scans** (224466 BNC-T, 224489 FIFO) to the Y-88 scale → absolute gain curves for setting HV.
- **Long run 224503**: SiPM MIP (per arm) and plastic triple MIP recheck.
- **Two significant discrepancies found** (flagged, need follow-up).
- Deliverable: **trigger threshold estimates** for SiPM sums and plastics.

Scripts `21--23f`; docs `HV_GAIN_Y88_ANALYSIS.md`, `report/y88_report.pdf`.

Y-88 absolute scale: both Compton edges, all three detectors







- Two edges on every plastic; measured 1612/699 ratio ≈ 2.4 (expect 2.31) *confirms* the energy assignment.
- Plastic gain **35–64 mV/MeVee** (HV raised for these runs).
- **First LIQ scale**: 32–37 mV/MeVee, consistent across arms.
- At 699 keVee all three detectors land within 21–27 mV per arm.

Discrepancy #1: plastic MIP is mis-calibrated by $\sim$40$\times$

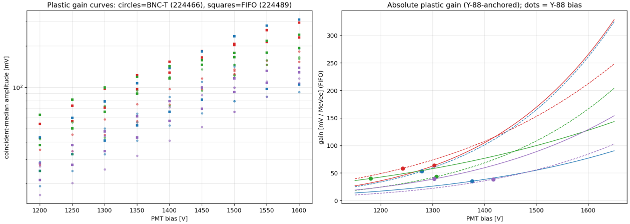
- The Y-88 line (known 699 keVee) gives plastic gain **35–64 mV/MeVee**.
- The old triples "MIP" (`19d`) assumed a **5.05 MeV** deposit and got 0.65–2.05 mV/MeV.
- $\Rightarrow$ the triples "MIP" peak really sits at $\sim$**130 keVee**, not 5 MeV — a $\sim$**40**$\times$ disagreement.
- Y-88 (BNC-T) and the 224489 MIP (FIFO) are otherwise *consistent* once the per-PMT FIFO factor (1.13–1.65) and HV are applied.

A real MIP in 2 cm plastic = 3.4 MeV (Landau MPV) $\rightarrow$ **120–217 mV** on the Y-88 scale.

The triples never find it (next slides).

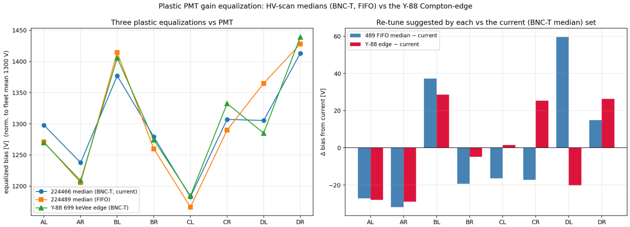
Trust Y-88; recheck the triples MIP with a dedicated low-rate / prompt-beam run.

Gain curves + Y-88 anchor → setting the HV



- Circles = BNC-T (224466), squares = FIFO (224489); differ by the per-PMT FIFO factor only.
- Y-88 pins the absolute mV/MeVee vs bias (right; dots = current bias).
- **Equalize + scale**: per-PMT bias to put 699 keVee at a chosen mV (table in `HV_GAIN_Y88_ANALYSIS.md`); e.g. 50 mV target ⇒ AL 1324 … DR 1559 V.

Do the three plastic equalizations agree?

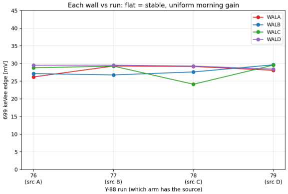


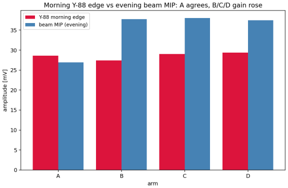
Three equalizations, all normalized to fleet-mean 1300 V:

- 224466 median (BNC-T) = **current fixed voltages**;
- 224489 median (FIFO); Y-88 699 keVee edge (BNC-T).

- **Same broad pattern** (CL lowest, DR highest); agree to $\sim \pm 25$ **V RMS**.
- Y-88 edge vs current: RMS **23 V**; FIFO median vs current: 31 V.
- Mostly they point the same way (AL/AR down, BL/DR up)...
- ... but **DL and CR disagree**: the MIP-tagged median and the 0.7 MeV edge want *opposite*

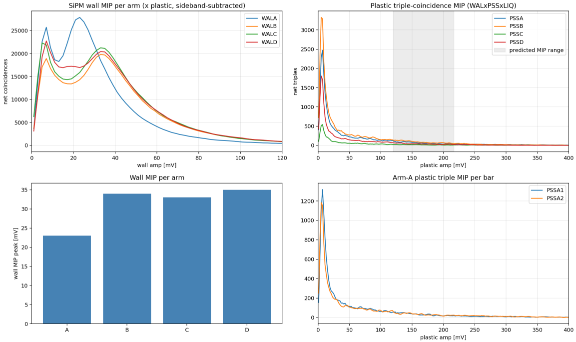
Discrepancy #2: SiPM MIP differs per arm (A low)





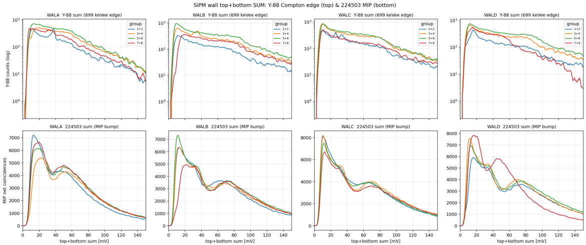
- Y-88 wall edge is **uniform & stable** across all 4 morning runs ($\sim$28 mV, left).
- But the beam MIP is **arm-A-low**: A $\approx$ MIP, B/C/D $\sim$1.4$\times$ higher (right).
- **Persists** in 224503 (next slide) $\Rightarrow$ not a transient.
- Uniform Y-88 (isotropic source) + non-uniform MIP (beam) $\Rightarrow$ **geometric** (per-arm beam incidence angle $\rightarrow$ path length), not a gain problem. *Confirm vs survey angles.*

Long run 224503: SiPM MIP clear, plastic MIP swamped



- **SiPM wall MIP** (top-left): clean bumps, WALA ∼23, B/C/D ∼33–35 mV — the arm-A gap persists.
- **Plastic triples** (top-right): net peaks at ∼5–7 mV, and is *flat* across the predicted 120–217 mV MIP band.
- Not statistics — the plastic's huge low-energy singles rate (∼20k hits/bunch) **swamps** the triple, and the late-tof cut drops prompt beam MIPs.
- ⇒ triples can't pull the plastic MIP from a high-rate beam run.

SiPM trigger thresholds (top+bottom SUM, 0.5×MIP)



Trigger fires on the top+bottom **sum** of each bar group [(1+2)(3+4)(5+6)(7+8)]. MIP bump of the sum (bottom row), threshold = half:

wall	MIP-sum [mV]	**thr [mV]**
WALA	50	**25**
WALB	70	**35**
WALC	67	**34**
WALD	72	**36**

Arm A lower (Discrepancy #2). Per-channel values in `calib/sipm_sum_thresholds.json`.

Plastic trigger thresholds (Y-88-anchored MIP estimate)

No usable triple MIP $\Rightarrow$ predict the plastic MIP from the **Y-88 scale**: MIP $= 3.4$ MeV (2 cm PVT) $\times$ gain.

PMT	AL	AR	BL	BR	CL	CR	DL	DR
predicted MIP [mV]	217	200	120	181	137	148	134	132
thr @1.7 MeV [mV]	**109**	**100**	**60**	**91**	**68**	**74**	**67**	**66**

- Threshold shown at 1.7 MeV ($\approx$ half the MIP); scales linearly — pick your energy.
- Per-channel because the current bias isn't perfectly equalized; equalize first (gain-curve slide) for a single common threshold ($\sim$122 mV at a 50 mV/699-keVee target).
- Plastic full scale $\sim$2000 mV, so huge headroom; threshold well clear of the $\sim$5 mV noise.

Recommended plastic HV: Y-88-edge equalization (FIFO)

Equalize each PMT's **699 keVee edge** (slid along its own power law) to a common target; target chosen to keep the **fleet-mean HV at 1301 V** (overall gain unchanged). Y-88 is BNC-T, so folded in the per-PMT FIFO factor.

CAEN 07._	AL .000	AR .001	BL .002	BR .003	CL .004	CR .005	DL .006	DR .007
current V	1303	1242	1376	1279	1180	1307	1303	1417
set V	**1237**	**1177**	**1440**	**1248**	**1214**	**1312**	**1331**	**1448**
change	−66	−65	+64	−31	+34	+5	+28	+31

- Arm A comes *down* (∼65 V): it has the highest FIFO factor (1.65/1.56), so it over-gains in FIFO. BL/DL/DR/CL come up.
- Range 1177–1448 V (HV scans ran to 1600 V fine).
- Machine-readable: `calib/plastic_hv_equalized_y88.json`. Target is a scale knob — raise it to shift all gains up together.

Summary & follow-ups

Thresholds to set now:

- SiPM sums: WALA **25**, WALB **35**, WALC **34**, WALD **36** mV (half the MIP-sum).
- Plastics: **60–110 mV** per PMT (1.7 MeV, Y-88 scale).

Two discrepancies to follow up:

1. Plastic triples "MIP" is at $\sim$130 keVee, $\sim$40$\times$ below a real 3.4 MeV MIP — the old `pss_mip_calib` energy scale is wrong; trust Y-88. Recheck triples with a low-rate / prompt-beam run.
2. SiPM MIP is $\sim$1.4$\times$ lower on arm A than B/C/D, persistently — likely geometric (beam angle). Confirm against the survey.

Figures `figures/21_y88/`; calibs `calib/y88_*.json`, `calib/sipm_sum_thresholds.json`, `calib/plastic_hv_gain_absolute.json`.